EXPERIMENT 5 — VACUUM CLEANER PROBLEM

AIM

To write a Python program to implement a **simple reflex agent for the Vacuum Cleaner problem**.

ALGORITHM

1. Define two rooms, **A and B**, as dirty initially.
2. Place the vacuum cleaner in room A.
3. Check the current room.
4. If the room is dirty, clean it.
5. Move the vacuum cleaner to the other room.
6. Repeat until both rooms are clean.
7. Display the cleaning process.
8. Stop.

PYTHON PROGRAM:

```
rooms = {'A': 'Dirty', 'B': 'Dirty'}

position = 'A'



while 'Dirty' in rooms.values():



    print("Vacuum at", position)



    if rooms[position] == 'Dirty':

        print("Cleaning", position)

        rooms[position] = 'Clean'



    position = 'B' if position == 'A' else 'A'



print("All rooms are clean.")
```

OUTPUT:

```
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Vacuum at A
Cleaning A
Vacuum at B
Cleaning B
All rooms are clean.
```

RESULT:

Thus, the **Vacuum Cleaner problem was successfully implemented using Python**.